

DUR E ADAN

Agentic AI & ML Engineer | 4x Gold Medalist

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PROFESSIONAL SUMMARY

Motivated undergraduate Agentic AI & ML Engineer at NUCES FAST Islamabad with hands-on experience in generative AI, retrieval-augmented generation (RAG), reinforcement learning, computer vision, and full-stack development. Proven ability to design and deploy end-to-end AI systems — from LLM-powered agentic applications to ROS2 robotics pipelines and production web apps. Experienced with PyTorch, TensorFlow, Hugging Face Transformers, LangChain, FAISS, and Gemini/OpenAI APIs. Quick learner with a strong foundation in prompt engineering, model fine-tuning, and environment-agent interaction, seeking to contribute to impactful AI engineering projects.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, HTML/CSS

Machine Learning: NumPy, Pandas, Scikit-learn, XGBoost, LightGBM

Deep Learning: PyTorch, TensorFlow, Transformers, Hugging Face Transformers

Generative AI: LLMs, Prompt Engineering, RAG, Embeddings, Fine-Tuning (LoRA/PEFT), LangChain, LangGraph, AI Agents, Multi-Agent Systems, MCP

Vector Databases: FAISS, ChromaDB

APIs: Gemini API, OpenAI API

Deployment: FastAPI, Streamlit, Docker

Research Tools: Jupyter Notebook, Google Colab, Weights & Biases (W&B), Hugging Face Datasets, Sentence Transformers

Robotics: ROS2 (Humble, Foxy), ROS1 (Noetic), Gymnasium, Stable Baselines, Gazebo, Rviz, Fusion 360

Web & Tools: React.js, MongoDB, JWT Authentication, REST APIs, Git, GitHub, Figma, Postman

PROJECTS

Campus Companion AI – RAG-Based Document Q&A Assistant

Python | Streamlit | FAISS | Gemini API | Sentence-Transformers

- Built a Retrieval-Augmented Generation (RAG) application that lets students upload university PDFs (handbooks, slides, syllabi, notices) and ask natural-language questions, with every answer grounded strictly in the uploaded content to prevent hallucination.
- Engineered a document pipeline that extracts and page-tracks PDF text, splits it into overlapping chunks, embeds them via a local Sentence-Transformers model or Google Gemini embeddings, and indexes them in a FAISS vector store for fast similarity search.
- Implemented retrieval-augmented answering with a tightly constrained Gemini prompt, returning responses with source citations (document name, page number) and a color-coded confidence score derived from retrieval similarity.
- Added Summary (short/detailed/bullet), MCQ generation (configurable count and difficulty), Simplify, and semantic Search modes, plus chat history export and embedding caching to avoid re-processing re-uploaded documents.

Hexapod Robot Control System

ROS2 | Gymnasium | Stable Baselines | Python

- Simulated a full hexapod robot using ROS2 Humble; implemented inverse kinematics for precise leg trajectory control.
- Designed reinforcement learning environments enabling autonomous locomotion learning via Stable Baselines agents.
- Integrated simulation, motion planning, and AI-based control into a complete end-to-end robotics pipeline with ROS2 nodes, topics, and publishers/subscribers.

Planora – Event Planner Web Application

React | Python | Flask | MongoDB | JWT

- Built full-stack event planning platform with React/Figma frontend, RESTful Python/Flask backend, and MongoDB Atlas.
- Implemented JWT authentication, Bcrypt password hashing, and Gmail SMTP notifications; validated all APIs with Postman.
- Managed full project lifecycle using Trello and Agile/Scrum methodologies for iterative delivery and sprint tracking.

PROFESSIONAL EXPERIENCE

AI Engineer Intern – Avancod Pvt Ltd

Jun 2025 – Aug 2025

Islamabad, Pakistan

- Developed computer vision pipelines using OpenCV; contributed to AI/ML model training and performance evaluation.
- Assisted in data preprocessing and end-to-end deployment of AI solutions for client-facing projects.

EDUCATION

B.S. Artificial Intelligence – NUCES FAST, Islamabad Campus

Aug 2024 – Present

F.Sc Pre-Engineering – Tameer-i-Wattan Public School & Colleges, Mansehra

Jul 2022 – Jul 2024

Grade: 92%

CERTIFICATIONS & ACHIEVEMENTS

- 4x Gold Medalist – Academic excellence recognition
- AI & Robotics Certificate – NUCES FAST Islamabad

LANGUAGES

- English — Professional Working Proficiency
- Urdu — Native Proficiency
- German — Elementary Proficiency
- Spanish — Elementary Proficiency
- Arabic — Elementary Proficiency